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REVIEW ARTICLE

Wildfire risk and insurance: research directions for policy scientists

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Abstract
Catastrophic wildfire is an increasingly familiar phenomenon on multiple continents. In the United States, concerns about uncontrolled, destructive wildfire have prompted some major insurance carriers to cease writing new policies or to non-renew existing policies. These trends affect not only policyholders, but also, vulnerable communities that already face multiple obstacles to securing property or renters insurance. This study reviews the social and behavioral sciences literatures on wildfire risk in the United States and insurance protection by homeowners. Three categories of research emerge from the review, namely, homeowner as rational actor, wildfire governance and risk management, and wildfire and social equity. There is abundant scholarship on determinants of homeowner decisions to manage wildfire risk by self-protecting or by purchasing insurance, but comparatively little research on the policy implications of shrinking markets for insurance. Policy research on the needs of underinsured and uninsured populations is also relatively undeveloped. Overlaying Lasswell's social process framework on the three dominant research themes, we find not only divergent research questions, models, and methods, but also, important differences in which stakeholders and stakeholder values are considered. There are opportunities for the different literatures to learn from one another, but also, to sharpen their focus on insurance as a scarce and uncertain resource amid climate change and as property development continues to expand in wildfire-prone areas.

Keywords Adaptive governance · Homeowners insurance · Moral hazard · Rational actor · Social equity · Social process · Social-ecological systems · Wildfire

Introduction

Wildfire is a natural feature of the life cycle and health of fire-dependent and fire-adapted ecosystems worldwide, and for millennia, fire was an intentional feature of indigenous land stewardship, including in the United States (Johnston et al., 2023; Kimmerer & Lake, 2001). Erratic and catastrophic wildfire, in contrast, reflects the legacy of eradicating

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natural occurrences and uses of fire in forest regeneration and land management. Large scale and uncontrolled wildfire events have surged in destructiveness in recent years in the United States (Hoover & Hanson, 2022) and are a source of growing concern in many other countries and regions, including Canada, southern Europe, and Australia (Canadell et al., 2021; Hazra & Gallagher, 2022). A combination of variables underscore wildfire risk management challenges in the United States, including a legacy of forest management policies and practices that encouraged wildfire suppression (enabling the build-up of flammable fuels in forested areas), increasing atmospheric temperatures, longer periods of low humidity, persistent drought, pest and pathogenic infestations and declining forest health, and expanding development inside the wildland urban interface (WUI) (Abatzoglou & Williams, 2016; Flannigan et al., 2000; Lucas et al., 2021).[1]

Wildfire risk: challenges for the insurance industry and for homeowners

Insurance companies are increasingly wary about wildfire peril (Jergler, 2023). Of the top-20 most destructive wildfires in California (measured by the number of structures destroyed), 15 have occurred since 2015 (CAL Fire, 2023). Citing wildfire risk as an important variable, two of the largest insurance carriers announced in 2023 they would cease writing new homeowners policies in California. Past losses from claim payouts and prospects of future losses informed these decisions. Across the nation, the industry reported more than $13 billion in losses from wildfire in 2020-primarily in the American West (Jacobs & Figuracion, 2021).

In recent years, the insurance commissioner in California adopted one-year moratoria preventing insurers from canceling or non-renewing policies for homeowners within or adjacent to declared state-of-emergency, wildfire-stricken zones. Many homes in these zones reside in the WUI, which raises questions about who pays for protection from destructive wildfire (Busby & Albers, 2010). California is only one among several Western states with large numbers of homes in the WUI. From 1990 to 2020, WUI-situated homes in states neighboring California increased by 46 percent. In this region, there were nearly 44 million units in the WUI by 2020, led by California with 5.1 million units (Gallagher Re, 2023: 5). Insurers' growing doubts about indemnifying homeowners and businesses in wildfire-prone states appear to affirm warnings made years earlier about imminent peril for insurers and policyholders amid climate change (Leggett, 1994; Flavin, 1996: 34).

Research by the risk analytics firm First Street Foundation finds that around 35.6 million properties in the United States (roughly one quarter of all properties) face increasing insurance prices due to climate change risks (including wildfire), decreased coverage, and property values that could decline due to real or perceived impacts of climate change (Picchi, 2023). Rising costs for insurance premiums are inspiring some homeowners to forgo insurance (Dagher, 2023). A survey conducted by the Insurance Information Institute and major reinsurer Munich Re found that 12 percent of homeowners were forgoing homeowners insurance. One in five respondents who did not purchase insurance had annual incomes of $40,000 or less (Insurance Information Institute & Munich Re, 2023).

[1] The increasing incidence and destructiveness of wildfire in the United States and its relationship to climate change has inspired Jennifer Francis of the Woodwell Climate Research Center to rethink the salience of "wild" in "wildfire." "We can't really call them wildfires anymore," Francis remarked to the Associated Press. "To some extent they're just not, they're not wild. They're not natural anymore. We are just making them more likely. We're making them more intense" (Borenstein & Walling, 2023).

In the United States, individuals and families buying a home with credit from a mortgage lender must secure homeowners insurance except in rare instances when the mortgage lender, itself, or a third party pays for the insurance policy. Homeowners policies vary in terms of coverages. The most popular type in the United States—the "HO-3" policy which applied to almost 78.3 percent of protected homes in 2020 generally cover perils like fire, lighting, wind, and hail (National Association of Insurance Commissioners, 2022). Insuring against flood risk and earthquake risk generally requires a separate policy. Major catastrophic fire events in the American West over the past several years have inspired some insurance companies to completely cease providing any type of homeowners insurance, regardless of the category of peril. As insurers exit the market, homeowners have increasing difficulty finding replacement insurance. A consequence is the sharp increase in the number of homeowners who settle for insurance-as-last-resort or FAIR Plan insurance. Nationwide, the aggregate value of all insurance in force in all FAIR Plans (excluding the value of coverages for liability and theft) was $837.3 billion in 2022 versus $324.8 billion in 2018—a nearly 158 percent increase during that period (Insurance Information Institute, 2021, 2022: 99).

To date, published research on insurance markets and wildfire risk management amid climate change has been a mostly unexplored subject in the social and behavioral sciences. This observation is borne out in the present study which examines publications that, at a minimum, make reference to wildfire (or wildland fire) and insurance protection by homeowners. This research enterprise is important insofar as risks from wildfire to people and property continue to mount and as traditional insurance markets are destabilized by catastrophic losses and prospects of future losses.

Methodology

This article considers the state of social and behavioral science research focusing on the challenge of insuring against wildfire risk in wildfire-prone areas of the United States. Keyword searches for articles referencing wildfire, wildland fire, and insurance were the basis for identifying articles to be reviewed. Titles emerging from keyword searches were combined with additional articles frequently cited in the relevant literatures, but not captured in keyword searches (Fig. 1).

The Science Citation Index (Web of Science/Web of Knowledge) platform, which includes publications from 1900 to the present (for this study, through May 19, 2023) was the foundation for a title and abstract search of published research on wildfire and insurance. Keyword searches were performed on the words "wildfire", "wildland fire", and "insurance". The Boolean operator "AND" was included between individual search terms so as to retrieve only publications containing all terms. To retrieve all permutation of "insurance", truncation (or stemming) was performed on the trunk word, "insur*". One hundred and seventy-six unique publications (inclusive of 11 books or edited books) were retrieved across searches; after elimination of duplicates, 166 records remained (appendix 1). Abstracts were reviewed for each article (or introductory paragraphs from chapters of authored or edited monographs). This initial screening of search results led to two rounds of additional culling. First, this review examined the insurance and risk management behavior of homeowners and renters. Hence, titles considering the insurance and risk decisions of other types of property owners—for example, owners of commercial timberland—were eliminated. Second, we limited the review to publications that implicated insurance

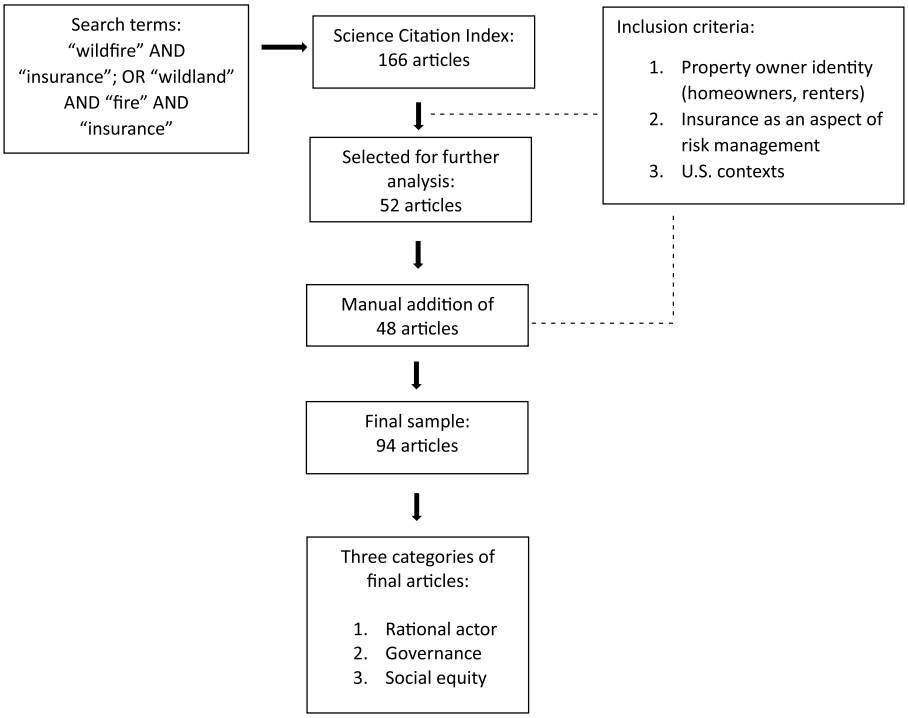

Fig. 1 Search strategy to identify articles on wildfire risk management and insurance. Source: review criteria after Snyder et al. (2016).

as an aspect of risk management—whether from the perspective of the insurance industry, government, or property owners and renters, themselves. Hence, we removed publications on wildfire management that mentioned insurance but were dedicated primarily to fire science, environmental science, or self-help, e.g., studies on fire suppression strategies, on the physical, ecological and economic impacts of fire on forests, and strategies for hardening homes. Third, articles addressing geographical contexts outside of the United States were eliminated, after Lambrou et al. (2023), so as to reduce variability in the legal-regulatory, institutional, and insurance market contexts examined. Fifty-two publications remained after subtraction of duplicates and following screening for policyholder type and geography. We manually added publications that consider homeowner wildfire risk management behavior that the Science Citation Index searches did not retrieve, including comparatively frequently cited publications (e.g., Collins, 2005; McCaffrey, 2004; McFarlane et al., 2011). Forty-two publications were manually added to the collection (appendix 2). These additional articles or monographs are generally frequently cited in the social and behavioral sciences literatures addressing wildfire, homeowner risk management, and insurance but lack the combination of terms entered into the Social Science Citation Index search platform. With the inclusion of the manually-added publications, a final tally of 94 publications were reviewed with attention to research questions, research design, materials and methods, results, and findings. The comparatively small set of research publications attending to homeowners insurance and wildfire risk affirms Hazra and Gallagher's observation that the field is understudied (2022:13), a view shared by Holmes et al., (2007: 144), and

affirmed quantitatively by Lucas et al., (2021:9) who found that, between 2009 and 2018, only three percent of research papers focusing on insurance and extreme weather primarily addressed wildfire.

Identifying categories of research on wildfire risk and homeowners insurance

Ninety-four publications from the compiled and culled literature were reviewed in their entirety by the author of the present study. A core theme across several publications in the final collection considers how homeowners, striving to protect their property from wildfire, respond to information and incentives. Several contributions adopt either theoretical, survey-based/econometric, or experimental (i.e., scenario-based and/or computational) methods to explore modeled and simulated behavior of subjects, and these articles are among the most frequently cited in the literature (Busby & Albers, 2010; McKee et al., 2004; Meldrum et al., 2014; Shafran, 2008; Talberth et al., 2006). Characteristic research themes include studies of decisions by property owners to insure or to mitigate risk, e.g., by making properties more resistant to fire. Various hypothetical and/or simulated conditions are considered. These include homeowners' choices and revealed preferences when disaster assistance is provided or public subsidies are made available to reduce risks at the neighborhood-level (Boomhower et al., 2023; Talberth et al., 2006). Other studies consider homeowners' perceptions of risk and past experiences with wildfire, and homeowners' observance of neighbors' insurance decisions and abatement behavior (Busby et al., 2013; Gardner et al., 1987; Holmes et al., 2012; Shafran, 2008). Many of these publications conceptualize homeowners as rational actors who weigh information and respond to incentives to buy insurance or to invest in risk-reduction with the goal of maximizing benefits at the lowest personal cost. There are 20 publications in this review with a predominantly rational actor perspective (appendix 3).

Another category of research on wildfire risk, insurance, and homeowner behavior conceptualizes homeowners as participants in multi-stakeholder and multi-sectoral institutions that adapt to wildfire risk. Disparate actors in organizational settings as varied as local, county, state, federal and Tribal government, private enterprise, community-based organizations, academicians and knowledge-based experts, and ordinary homeowners have a shared interest in protecting livelihoods, property, and other valued assets from destructive wildfire (Abrams et al., 2021; Bixler et al., 2023; Kirschner et al., 2023). Relationships, trust-building, mutual understanding of roles and responsibilities, flows of information, and planning and implementation of wildfire risk management strategies are key variables in these studies. Providers of insurance are participants in governance institutions, and insurance contracts are tools that networked actors rely on to manage risk. Twelve publications in this review primarily consider wildfire risk as a governance challenge (appendix 3).

The third, dominant theme in the literature review focuses on vulnerable people and communities grappling with wildfire risk. These populations are at-risk from wildfire not merely because of their physical proximity to wildfire-prone areas but because they are underserved and tend to be beyond the reach of policies and programs that protect people from unplanned and uncontrolled wildfire. Lower income homeowners, lower income renters, seasonal workers, and elderly and disabled people are frequent subjects in this category of research, and gaps in policies and programs to protect these populations are recurrent subjects of inquiry. A third and final category of research—in this case, on wildfire and vulnerable communities—considers elevated risks and unequal outcomes for marginalized populations. We adopt "social equity" as a shorthand for this category. Fifteen

publications in this review primarily document differences in access and differential outcomes for people and communities that are, at once, susceptible to wildfire risk and are underserved—including in their ability to secure high quality property insurance against wildfire (appendix 3).

Not all retained and reviewed publications fall into one of the three dominant categories ($n = 48$; see appendix 3). There are, for example, several publications addressing technical subjects that are not addressed in other publications (see, e.g., Troy & Romm, 2007; Frazier, 2021; Sterk, 2023) or that consider insurance only tangentially (e.g., Plantinga et al., 2022). Nevertheless, these titles were reviewed in their entirety, and as relevant, their findings are considered in this article.

Cross-walking the categories

We create a framework to capture a broad array of key variables for scholars of wildfire risk management and insurance to consider by cataloguing and comparing key actors invoked in the three relevant literatures, their perspectives, and the strategies they use to achieve their goals. Homeowners deciding whether to insure or to reduce wildfire risk on or around their homes, are the main subjects in the rational actor literature. In contrast, homeowners are one among a diverse set of networked actors in the governance literature. Underinsured and uninsured occupants of dwellings—including many who have no equity stake in properties they live in—dominate the social equity literature. Harold D. Lasswell's social process, with its attention to people, their demands and expectations, and the outcomes and lasting effects of their cumulative decisions is a fitting tool to compare the wide range of social and behavioral science literature on wildfire risk management and insurance. In the synthesis section of the present article, this is the framework we adopted to compare and contrast the three categories of reviewed research. The synthesis is preceded by the findings of the literature review, beginning with rational actor approaches, followed by scholarship on governance, and finally, research on social equity.

Three categories of literature

The principal research concerns, methods, and findings of the reviewed literature begins with the rational actor category followed by contributions on governance and on social equity.

Homeowner as rational actor

The homeowner, as rational actor, weighs strategies to manage wildfire risks. Those strategies include mitigating (or abating) risk (such as removing flammable vegetation near the home), buying insurance, or some combination of both. Economists have examined the insurance versus mitigation trade-off using theoretical models and experimental and econometric methods for several decades. Of longstanding interest is the problem of moral hazard (Boomhower et al., 2023; Kunreuther & Michel-Kerjan, 2007; McKee, et al., 2004; Talberth et al., 2006). Moral hazard posits that an insured party, recognizing they are indemnified from loss, will engage in risky behavior to the detriment of the insurer (Kotowitz, 1989). In the context of wildfire, moral hazard might take the form of a policyholder who underinvests in wildfire mitigation (Brenkert-Smith et al., 2015) or

who underestimates potential losses due to wildfire (McNeill et al., 2013). From a rational choice (or rational actor) perspective, the insurance contract interferes with an otherwise straightforward balancing of costs and benefits by the homeowner. However, multiple empirical studies reveal homeowners willing to invest in private abatement to avoid losses including when properties are fully insured (Fried et al., 1999; Simmons & Kruse, 2000; Simmons et al., 2002; Winter & Fried, 2001). Higher-than-expected willingness to pay for abatement may occur because the homeowner values intangible losses that may not be covered by insurance (Fried et al., 1999; Simmons et al., 2002). Combining willingness to pay, survey research, and an interactive computer program, Talberth et al., (2006: 209) found that 94 percent of subjects were willing to spend money on mitigation and insurance, even when insurance covered 100% of tangible losses. The authors conjecture that subjects chose to invest in abatement and insurance as part of a package, and that "amenities… the quintessential nonreimbursable aspect of home ownership" was a significant factor in homeowners' overall willingness to pay. Briefly, empirical research challenges the basic tenets of the moral hazard conundrum—at least as it applies to purchasing insurance and/or opting to abate. Variables that influence decisions to simultaneously mitigate and/or insure include homeowners' perceptions of risk and their expectation that government will help protect property. Risk perception and government assistance are two major sub-themes in the rational actor literature on wildfire risk and insurance. Each is considered, below.

1.1. Risk perception and rational choice

Econometric and psychometric research on wildfire risk and homeowner behavior is well-developed (Ghasemi et al., 2020; Liu & Jiao, 2018) though this literature seldom features homeowners insurance as the dependent variable. Of abiding interest is whether and to what extent homeowners' perceptions of the probability of wildfire (Gardner et al., 1987; Martin et al., 2009), or alternately, of the consequences of wildfire (Boakye et al., 2022; Lambrou et al., 2023; McNeill et al., 2013) spur self-protective action (Donovan et al., 2007; McFarlane et al., 2011; McGee et al., 2009; Warziniack et al., 2019). Past experience with wildfire and willingness to pay to avert future wildfire is a related theme in this literature (Black et al., 2023; Champ & Brenkert-Smith, 2016; McGee et al., 2009). Brenkert-Smith et al., 2012 found that households were more likely to mitigate risk following direct experience with wildfire. Similarly, Ghasemi et al. (2020) found that community-based wildfire-reduction activities were popular with homeowners who had firsthand experience with wildfire. In contrast, Schulte and Miller (2010) found no significant relationship between past experience with wildfire and subsequent investment in mitigation.

2.2. Government action and homeowner behavior

Several experts consider whether and to what extent government transfers and other forms of public assistance in wildfire management affect homeowners' decisions to mitigate, to insure against risk, or both. Types of government policies and actions include: (a) provision of resources to suppress wildfire risk; (b) promise of disaster relief; (c) provision of funds for private- and/or local-level abatement; (d) permissiveness toward development in wildfire-prone areas; and (e) enabling of insurance of last resort (Table 1).

Each study in Table 1 considers, but is not limited to, clarifying whether the provision of government resources disincentivizes wildfire mitigation behavior by homeowners or affects homeowners' insurance decisions. A generalized finding, across these contributions,

Table 1 Wildfire risk, government intervention, and homeowner behavior

Categories of government policies and actions	Relevant literature
a. provide fire suppression services	Busby & Albers (2010) Busby et al (2013) Champ et al. (2020) Cohn et al. (2008) Collins (2005) Donovan & Brown (2007) Holmes et al. (2007) McCaffrey (2004) McFarlane et al. (2011) McKee (2004) Prante et al. (2011) Pyne (2008) USDOI & USDA (1995) Vogt & Cindrity (2003) Winter & Fried (2001) Yoder & Blatner (2004)
b. promise disaster relief	Baylis & Boomhower (2023) Boomhower et al. (2023) Holmes et al. (2007) McKee et al. (2004)
c. provide funds for private and/or local abatement	Berrens et al. (2010) Lankoande et al. (2005) McKee et al. (2004) Meldrum et al. (2014) USDOI & USDA (1995) Talberth et al. (2006)
d. allow property development in wildfire-prone areas	Alavalapati et al. (2005) Colburn (2008) Dixon (2018) Dombeck et al. (2004) Philson et al. (2021) Pyne (2008)
e. enable insurance of last resort	Gallagher & Hazra (2020) Hazra & Gallagher (2022) Troy (2007)

is that research on homeowners' self-protection behavior is far more common than on insurance decisions. Second, as to whether government intervention has a significant effect, the results are mixed, reflecting the studies' greatly varying methods and data sources, ranging from theoretical models to contingent valuation and survey-based research, regression analysis, and machine learning. There is one humanistic perspective (Pyne, 2008), three agency perspectives (Dombeck et al., 2004; USDOI & USDA, 1995; Vogt & Cindrity, 2002), and one law review (Coburn et al., 2008) among the publications considered in Table 1.

The effects of government-sponsored wildfire suppression activities receive the most attention (category a in Table 1). When fuels are treated or forests are thinned on adjacent public lands, and/or when firefighting services are expected by homeowners, incentives for household-level abatement decrease (Busby & Albers, 2010; Cohn et al., 2008; Collins, 2005; Donovan & Brown, 2007; Holmes et al., 2007; USDOI & USDA, 1995; Vogt & Cindrity, 2002; Winter & Fried, 2000; Yoder & Blatner, 2004). A notable exception is Champ

et al., (2020: 309). For publicly-funded thinning and prescribed burns on nearby public lands, the hypothesis is that these activities would dampen homeowners' interest in private abatement since the state is bearing the cost of risk reduction and homeowners can free ride. Shafran (2008) found the opposite effect with homeowners creating less defensible space if the neighboring property was untreated public land. In a computerized experiment, Prante et al. (2011) found that subjects spent more on mitigation, and were more likely to mitigate, when collective information on neighbors' mitigation expenditures was provided. Holmes et al. (2007) is notable for considering not only private abatement behavior but also private insurance decisions amid government-sponsored wildfire suppression activities. Invoking economic theory, they reason that government action reduces homeowner incentives to purchase insurance as well as to self-protect.[2]

Holmes et al. (2007) is among a small set of empirical contributions that consider how disaster assistance affects homeowner insurance decisions (category b in Table 1). The promise of disaster relief, they contend, is often deemed an entitlement by homeowners in wildfire-prone areas, and they propose that "…if property owners believed that disaster relief would not be forthcoming from the federal government unless effective risk-mitigation measures had been implemented beforehand, then their incentives for self-protection would be enhanced" (2007: 153). Their concern is affirmed in models by Baylis and Boomhower (2023) and Boomhower et al. (2023). In contrast, McKee et al. (2004) found that subjects opt for *more* insurance coverage when disaster relief is available—a counterintuitive result from a moral hazard perspective. The authors propose that the "warning effect" of an announcement of public disaster aid (e.g., by the Federal Emergency Management Agency) could motivate homeowners to purchase insurance, or conceivably, homeowners expect aid eligibility to be contingent on having an insurance policy. In an experiment by McKee et al. (2004), moral hazard was not evident in homeowners' insurance purchases, but information about the availability of government disaster aid did lead subjects to reduce private abatement efforts.

There are a variety of federal and state government programs that cost-share abatement activities at neighborhood- and local-levels as well as wildfire resilience programs that assist individual homeowners, i.e., on private land (category c in Table 1). This type of government assistance is different from the aid described in category a, Table 1 which involves publicly-funded fire suppression on *public* lands. Several contributions consider whether homeowners' private investment in abatement is dampened when government steps in to help (Lankoande et al., 2005; McKee et al., 2004; Talberth et al., 2006; USDOI & USDA, 1995), but there is much less research on how public subsidies for private abatement affect insurance decisions. McKee et al. (2004) is unique in considering whether public transfers for abatement affect homeowners' decisions to buy insurance. They found no moral hazard; insurance purchases were unchanged when government aid was available.

A fourth form of government activity addresses lax zoning ordinances, permissive building codes, and related regulatory gaps in wildfire-prone areas (category d in Table 1). Several studies point to lax property development policies as primary drivers of sprawl

[2] In addition to studies of homeowner mitigation behavior in response to abatement activity on adjacent public lands is homeowner behavior vis-a-vis abatement on neighboring *private* land. McFarlane et al. (2011) found that private mitigation was positively correlated with neighbors' mitigation activity. Berrens et al. (2010) obtained a similar result. Steinberg (2011) found mixed outcomes in local-level efforts to reduce wildfire risks to neighborhoods and to individual homes under the mantle of Firewise Communities/USA® programs.

in wildfire-prone areas, including the WUI (Alavalapati et al., 2005; Coburn et al., 2008; Dombeck et al., 2004; Philson et al., 2021). To the extent they consider insurance as a variable, critics of sprawl in wildfire-prone areas predict a crisis for insurance companies and for state-mandated insurance plans (Pyne, 2008) or contend that insurance rates are too low to discourage development in high-risk areas (Dixon et al., 2018).

Insurance of last resort is a final category (e in Table 1) of government support that might affect homeowners' wildfire management behavior. It is conjectured that the provision of state-mandated last resort insurance might lead to reduced private abatement as homeowners expect to be indemnified against losses. State-mandated Fair Access to Insurance Requirements or FAIR Plans provide insurance to homeowners living in very high risk areas (e.g., prone to natural disaster or high-frequency crime) who are denied insurance in the standard market. In several states, in order to secure a license to write standard insurance policies, carriers must also be part of the state's last resort plan. Hazra and Gallagher (2022) found that hazard class (e.g., very high fire hazard severity zone designation) had a statistically significant impact on FAIR Plan insurance premiums. However, the importance of location hazard was small relative to various housing attribute factors. Moreover, the effect of wildfire risk on insurance premiums increased at a diminishing rate. The failure of premiums to reflect higher costs in higher-risk zones enabled perverse incentives for home construction in these areas (Hazra & Gallagher, 2022).

Wildfire governance and risk management

In the rational actor literature, government resources or actions, like thinning on public land or fire suppression, are salient insofar as they affect homeowners' choices about risk management. In contrast, the concept of *governance* is more nuanced and multi-layered, and involves not merely assets provided by public authorities, but the formation of complex networks of like-minded actors and institutions dedicated to wildfire adaptation and resilience. "Governance", Folke et al. write (2005: 441).

> … connects individuals, organizations, agencies, and institutions at multiple organizational levels. Key persons provide leadership, trust, vision, meaning, and they help transform management organizations toward a learning environment. Adaptive governance systems often self-organize as social networks with teams and actor groups that draw on various knowledge systems and experiences for the development of a common understanding and policies.

Decisions made by homeowners to purchase insurance, abate, or take advantage of government resources for wildfire risk management are embedded in these broader sets of rule-based institutions and information networks. Scholarship on the formation and function of governance institutions is abundant (see Kirschner et al., 2023:appendix 2), addressing subjects ranging from efforts to define particular forms of adaptive governance and the variables affecting their performance and effectiveness (Bixler et al., 2023; Bourceret et al., 2021; Cumming et al., 2006; Guerro et al., 2013; Kirschner et al., 2023; Termeer et al., 2010) to research on coordinated actors managing environmental challenges or services in specific geographical and/or social-ecological contexts (Abrams et al., 2021; Bergmann & Bliss, 2004; Epanchin-Niell et al., 2010; Nelson et al., 2008; Tait & Brunson, 2021). Many governance studies consider one or more of four main themes: (1) complexity; (2) collaboration and co-production; (3) path dependence; and (4) adaptation and resiliency. We summarize major insights from each theme, below.

1.1. Complexity

The governance literature frequently categorizes unplanned wildfires as "social-ecological" (or socio-ecological) phenomena. Erratic wildfires often involve geographical, topographical, and ecological complexity with disparate institutions for crisis management (Abrams et al., 2021; Steelman, 2016). Apart from their volatile nature, uncontrolled wildfires tend to pose institutional challenges, such as complex property rights regimes, authority fragmentation, and authorities working in siloes (Steelman, 2016). Information sharing and collaboration are potential antidotes. Governance experts like Daniels and Walters (1996) and Suškevičs et al. (2018) find inclusiveness, transparency, and accountability to be especially important factors in successful and enduring regimes for forest resource management, including wildfire management.

Stakeholders with financial and insurance risk expertise are potential collaborators in wildfire governance systems. Reliance on different participants' specialized knowledge and skills can enable the entire network to succeed by learning and adapting. Different knowledge-based actors become conversant with different aspects of the problem, and then, bring their expertise together as problem-solving collaborators. Hence, as Kunreuther (2000) observes, authorities who enforce safety standards, like building code inspectors, can combine their knowledge with the risk-forecasting and actuarial expertise of banks and insurance companies to promote investment in cost-effective risk-reduction measures. But whether property insurance, itself, reduces or alternately increases complexity in wildfire governance regimes is an open question in the governance literature. Consider, for example, a hypothetical zoning rule meant to incentivize investment in fire-resistant building materials for homes in the WUI. Insurance could be used as leverage for this policy, i.e., if owners of older properties fail to retrofit their homes, they could be required to buy insurance. On the other hand, promoting policies to provide affordable insurance for homeowners in the WUI may inadvertently lead to expansion of construction in wildfire-prone areas (Paveglio et al., 2018).

2.2. Collaboration and co-production

Precisely because of their complexity, social-ecological systems require collaborative partnerships if these systems are to be successfully managed. No single actor can manage the complexities of planning for, responding to, and recovering from erratic wildfire. Collaboration is needed across political boundaries, between state and non-state actors, and between public agencies and ordinary citizens (Davis et al., 2021; Muller & Yin, 2010; Steelman et al., 2021). At the heart of collaborative networks for wildfire management, according to Abrams et al. (2016) are local-level actors who participate in decision-making institutions. Indeed, the U.S. federal government has urged that wildfire management and adaptation can only succeed when actors and organizations at multiple levels, including the local level, are deeply engaged. This multi-level, cross-sectoral approach was the motivation for the creation of the 2014 National Cohesive Wildland Fire Management Strategy (or Comprehensive Strategy) which united government agencies from federal to local levels and non-governmental organizations to promote resilient landscapes, fire-adapted communities, and safe and effective wildfire response (USDOI & USDA, 2014). However, it has been observed that among the multi-stakeholder commissions that oversee implementation of the Cohesive Strategy, federal agencies are overrepresented, creating imbalances of power, rigid approaches to planning, and side-lining of Tribal and local communities'

placed-based knowledge of wildfire and wildfire management (De Abreu, 2022; Steelman et al., 2019). Similarly, Miller et al. (2008) report on the failure of government agencies to tap local knowledge of wildfire. Hence, responsible agencies lack integrative knowledge, hampering creation of more resilient fire management strategies.

2. Path dependence

The idea of path dependence posits that past decisions, practices, customs, and habits are reflected in efforts to revise policy solutions (Wilsford, 1994) As Douglass North (1990) posits, path dependency is a process that restricts future choices. In the context of wildfire risk management, decisions and practices adopted years ago complicate or constrain strategies for governing areas at-risk of destructive wildfire (Cheng et al., 2010; Johnson et al., 2023; Kirschner et al., 2023). Path dependence has important implications for rule-making by higher-level authorities when protected areas and wildfire risk zones are designated, and by extension, when insurance companies adapt to these designations. In California, very high fire hazard severity zones (VHFHSZ) have been designated throughout the state following adoption of new rules in 2007 (Conservation Biology Institute 2022). Frequently, VHFHSZ have been assigned to areas that, in decades past, were designated for protection. Higher-income individuals and families built homes in or near these areas, such as in Oakland Hills and Mill Valley, to take advantage of spectacular views, recreation opportunities, and proximity to nature (Donovan & Brown, 2007). In these same zones are lower income homeowners whose families have resided continuously for generations and well before the VHFHSZ and protected areas designations. Decisions made by insurance companies to cancel or non-renew insurance policies impact wealthier as well as less-wealthy homeowners in these areas, though better-off households have more insurance options. Path dependence is salient in this context insofar as complex geographical mosaics of insured and underinsured properties can result from decades-old decisions about protected areas and the assignment of wildfire risk zones. Residents' uneven access to fast-changing insurance markets are another path-dependent result. To understand the legacy of which participants are at-risk and why, insurance companies and regulators are well-advised to consider historical contingencies and knock-on effects of past policy decisions and practices.

3.3. Adaptation and resiliency

Social ecological systems that are successful tend to adapt to changing environmental and institutional conditions (Folke et al., 2010). They do so by learning—drawing on experience and knowledge (Berkes, et al., 2003). Networks that develop and deploy new knowledge and that refine plans and strategies for problem-solving are able to overcome path dependencies. A collaboration between the Nature Conservancy and insurance broker Willis Towers Watson provides an illustration. The two organizations co-developed an adaptation to wildfire risk while reducing hypothetical insurance premium costs for homeowners in the Tahoe National Forest in Placer County, California. Application of ecological forestry (which includes thinning and prescribed fire to reduce ladder fuels) generated savings of around $21.1 million in aggregate annual home insurance premiums per year (The Nature Conservancy & Willis Towers Watson 2021). In this case, "parametric insurance" rather than typical indemnity insurance was the basis for determining savings to property owners. Parametric insurance pays out when a particular "parameter" is met or exceeded, such as acres burned (The Nature Conservancy and Willis Tower Watson 2021: 5). This

approach diverges from indemnity insurance whereby the claimant must prove that losses were incurred due to damage or destruction of an insured asset. Parametric insurance products are not new, but their application to wildfire risk management are exceptional. Payouts from this type of insurance are pre-determined, posing less uncertainty to the payer (i.e., the insurer). Insofar as insurance companies and authorities that regulate them (namely, insurance commissioners) are participants in adaptive networks, parametric insurance could emerge as important tool for social ecological adaptation and resilience (Hedberg & Franco, 2022).

Wildfire and vulnerable populations

Poverty, disability, linguistic isolation, lower levels of education, and other challenges affect vulnerable populations' capacity to manage wildfire risk, including securing adequate insurance (Collins & Bolin, 2009; Collins, 2008a; Ojerio et al., 2011; Simon & Dooling, 2013; Wigtil et al., 2016). Whether disaster relief or government-sponsored abatement funds dampen homeowners' incentive to buy insurance may be less relevant when the subjects of interest are struggling to pay for basic needs. Or consider renter behavior: For the purpose of preserving property value, renters may have low intrinsic interest in private abatement. But more saliently, they are disadvantaged when landlords and rental agreements fail to protect renters from wildfire risk. As Collins (2008b) point out, renters are often legally prohibited from making physical changes to properties to make them safer, and renters' vulnerability is correlated with other risk factors such as low-income status and lack of renters insurance.[3]

Literature on wildfire risk and vulnerable populations consider questions of equity and environmental justice (Auer, 2021; Auer & Hexamer, 2022; Canes, 2019; Davies et al., 2018; Liao et al., 2022; Méndez et al., 2018). Research highlights access to resources (Lambrou et al., 2023; Sánchez et al. 2022; Wigtil et al., 2016; Simon & Dooling, 2013) more often than to moral hazard or revealed preferences involving private mitigation or purchase of insurance. Justice, and the related theme of equity, are overarching concerns. Schinko et al. (2023) provide perhaps the most nuanced approach to understanding justice in wildfire risk management, by defining and mapping aspects of distributional, procedural, and representative justice across various stages of wildfire management, from prevention to preparedness, response, and recovery and adaptation. Insurance contracts, they note, have important justice implications depending on whether applicable rules and regulations or industry practices include provisions for "solidarity" in pricing, so as to subsidize poorer

[3] In 2022, 45.2 million homes were renter-occupied in the United States versus 129.9 million owner-occupied homes (U.S. Census, 2023a). More than 10 million renters live in areas prone to natural disasters, including (but not limited to) hurricanes, earthquakes, tornados, and fire (Joint Center for Housing Studies of Harvard University, 2020: 37). Drawing on data from the American Community Survey, the Joint Center for Housing Studies of Harvard University found that between 2015 and 2017, 324,000 renters were displaced by natural disasters in the United States and 500,000 rental units required substantial repairs after disasters struck (Joint Center for Housing Studies of Harvard University, 2020: 37). Apart from challenges renters may face, such as prohibitions in lease agreements that could help protect renters in wildfire-prone areas, there are other household financial stressors that impair renters' ability to be safer from wildfire. As a very rough proxy for renters in WUI areas, consider that over 39% of renters in rural areas in the United States in 2018 were cost-burdened, meaning they paid more than 30% of income for rent and utilities (Joint Center for Housing Studies of Harvard University, 2020: 28). For comparable data, see the U.S. Census monthly Household Pulse Survey, including indicators such as "Household currently caught up on rent payments" (U.S. Census., 2023b).

households. Equity is the core concern of Lynn and Gerlitz (2006) who found many lower income households in wildland areas in Oregon and Washington residing outside WUI zones serviced by federal disaster relief programs. Liao et al. (2022) considered whether homeowner insurance nonrenewal rates were affected by social vulnerability factors in several hundred zip codes in California. While they did not uncover disproportionate rates of nonrenewal among vulnerable populations generally, they did find hot spots of nonrenewal among lower income households in various regions of the state. Lambrou et al. (2023) examined a vast set of social vulnerability characteristics inclusive of but not limited to income, race, citizenship, female-headed household, mobile home occupancy, multi-family home, overcrowding/people per unit, and lack of personal transportation. Their conceptual model of vulnerability emphasized the location of power in wildfire governance networks, the preparedness of communities marginalized by income and race, the importance of family as sources of information about wildfire risk, and place attachments as key determinants of resilience. They write,

> People and agencies in positions of power that make decisions which affect all households in a community are not impartial, and are likely to represent the interests of some groups over others, often disguising the fact that disasters are social and not natural (Lambrou et al., 2023:9).

Researchers can and should include insurance companies, insurance regulators, and state legislatures among "people and agencies in positions of power" as they examine the differential impacts of insurers' decisions to cancel or non-renew insurance or cease writing new policies in particular zones or in entire states. Early indications that some powerful institutions are recognizing the plight of vulnerable populations include the California Commissioner of Insurance's decision to impose year-long moratoria on insurance non-renewal in areas affected by, or near to, catastrophic wildfire—a decision influenced by the plight of people "…like retirees…on fixed incomes, working people, and those pushed out of the urban core" (U.S. House of Representatives, 2022a). Support for disadvantaged communities is also featured in FEMA's Building Resilient Infrastructure and Communities grant program. The program provides a 90% federal cost share (versus the standard 75% cost share) for Economically Disadvantaged Rural Communities (FEMA, 2023).

Synthesis

A review of the social and behavioral sciences literatures on wildfire management, homeowner behavior, and insurance reveals major differences in problem statements and in methods for illuminating challenges and choices faced by insured and uninsured parties. Studies exploring moral hazard, for example, generally do not consider socioeconomic conditions that affect choices to mitigate or to insure. As Lucas et al., (2021:17) observe,

> It is important to recognize that what are often presented as technical and actuarial decisions made in applying insurance obscure the political and moral logics embedded within them. Governments should ensure that insurance logics do not disadvantage the least privileged members of society, even if such protection involves regulation and unwelcome intervention in the insurance industry.

As it becomes increasingly challenging for people to navigate markets for insurance against wildfire loss, there is a pressing need for researchers to carefully consider the baseline

conditions of participants in wildfire management, their value demands, and their strategies for goal attainment (Nowell et al., 2022).

Identification of key actors and their values, strategies, and preferred outcomes are first order questions for policy scientists to grapple with. The mapping of participants, their perspectives, and the outcomes and effects of strategies they pursue is the domain of the social process framework in Harold D. Lasswell's policy sciences (Clark, 2002; Lasswell, 1971; Lasswell & McDougal, 1992). Overlain on the three categories of wildfire risk literatures, the social process uncovers not merely the technical considerations people make when managing risk, but also salient norms and social-cultural variables that shape all human interactions, such as the extent to which people show respect for government authority or obey commands by a head of household (Brunner, 2004; Clark, 2002; Lasswell, 1971). Nowell et al. (2022), in their critique of the co-management literature on multi-jurisdictional responses to wildfire, affirm the value of the social process for elucidating the many, varied, socially-embedded dimensions of wildfire management. We do not address Nowell et al.'s query as to whether the time-tested value of the policy sciences preclude the need for a science of co-management (Nowell et al., 2022: 532). Rather, we use Lasswell's framework to catalogue and explore the social dimensions of wildfire risk and insurance, with the literatures on rational actor, wildfire governance, and social equity as data sources.

Table 2 crosswalks the three literatures. Essential participants, the assets or resources they use, dominant strategies they adopt, and preferred end states they strive for, collectively, are presented (Lasswell & McDougal, 1992:150).

The crosswalk reveals conspicuous differences in which participants are emphasized in the respective literatures, the value assets participants use to achieve their goals, and preferred end states. Consider, for example, that vulnerable populations are undifferentiated—or entirely unrecognized—in the rational actor literature.[4] Segmentation of subjects in the social equity literature underscores particular value assets or deprivations of particular populations. Renters' key value assets are among the most abbreviated in the table. This is consistent with Davies et al. (2018) who observe that federal fire recovery funds tend not be available to renters, and it reflects the higher probability that renters are geographically isolated from family or friendship networks.

The crosswalk also points to major differences in the respective literatures' identification of government participants and functions. While the rational actor and governance literatures make reference to many of the same levels of government involved in wildfire management, the organizational interests and roles of these actors are often simplified in rational actor models. Theoretical treatments, for example, Boomhower et al. (2023), are spare in terms of actors and activities considered. "Government" is not differentiated by jurisdiction or level, and the key activity is government disaster relief. A comparable, parsimonious model is adopted by Lankoande et al. (2005). In contrast, the governance literature is more expansive about levels of government, agency responsibilities, and conditions for collaboration and coordination (e.g., Steelman, 2016).

In successful wildfire governance regimes, federal, state, local and Tribal governmental agencies are networked actors whose success depends on shared information, systems of accountability, and nimble coordination across temporal scales (Essen et al., 2023). In the social equity literature, key agencies operate at the federal and state levels providing disaster relief and wildfire resilience resources, or at the local level, providing social services

[4] Notable exceptions are Lucas et al. (2021) and Hazra and Gallagher (2022).

(Cutter et al., 2008; Fu, 2013). Governments also engage in intelligence gathering and promotional activities (U.S. House of Representatives, 2022b) though this tends not to be a major topic in any of the literatures.

Table 2 Crosswalk of social process categories and wildfire risk literatures

Social process categories / Wildfire literatures	Prominent participants		Key value assets	Principal strategies	Preferred outcomes
Rational actor	Homeowners		Equity in home; disposable income for insurance and/or mitigation (wealth)	Self-interest; maximization of economic returns	Well-protected homes and communities reflecting Pareto optimal allocations of private funds toward abatement and insurance purchases, supported by well-designed government policies
	Insurers		Expert knowledge (skill); wealth	Actuarial science; risk assessment; artificial intelligence; augmented reality	
	Government	Federal	Funds for disaster relief and wildfire resilience (wealth) and capacity for wildfire mitigation and suppression (skill)	Government transfers for abatement on public and private property; disaster relief funds; fuel management on public lands; public awareness campaigns	
		State	Regulatory power, e.g., insurance of last resort; building codes; funds and capacity for wildfire mitigation and suppression (wealth, skill)	Subsidization of and/or regulation of insurance of last resort; applicant for federal designation of disaster zones, application for federal disaster and Resilience funds	
		County	Zoning, land-use regulation, and building code authority (power); firefighting services (wealth, skill)	Wildfire zone determination process; allocation of firefighting resources; Applicant for federal resilience funds	
		Local	Zoning, land-use regulation, building code, vegetation management authority (power); firefighting services (wealth, skill)	Wildfire zone determination process; allocation of firefighting resources; formulation and adoption of building codes and construction standards	
Governance	Homeowner		Information from multi-scale actors on wildfire mitigation resources (enlightenment)	Information-sharing with same-actor networks (e.g., intergovernmental and across jurisdictions) and within networks composed of state- and non-state actors (e.g., creators of Community Wildfire Protection Plans); collaborative problem identification; participatory rule-setting; coordination of rules and procedures across levels, scales, and jurisdictions; consultation on aligning territorial and ecological boundaries; collaborative planning; joint monitoring and enforcement; contractual relationships with knowledge-based experts; resource transfers from government to multi-stakeholder planning and implementation teams	Effective wildfire resilience at community, local, state, national, international, and global scales, enabled by successful collaboration of actors across different jurisdictions and scales
	Government	International organizations	Treaty, compact, directives, soft-law (power)		
		Federal	Collaboration (skill, respect); legal-regulatory (power); public land management (power, wealth, wellbeing); resources for fire suppression, abatement, disaster relief (power, wealth, wellbeing)		
		Tribal	Legal-regulatory (power, respect); collaboration (skill, respect); indigenous knowledge (enlightenment, skill)		
		state	Collaboration (skill, respect); legal-regulatory (power); resources for managing wildfire (wealth, power, skill)		
		county	Collaboration (skill, respect); legal-regulatory (power); resources for managing wildfire and taxation for these services (power, wealth, skill)		
		Local	Collaboration (skill, respect); legal-regulatory (power); resources for managing wildfire and taxation for these services (power, wealth, skill)		
	Non-state actors	Advocates for insured	Knowledge (enlightenment); voice (power); collaboration (skill, respect)		
		Advocates for vulnerable populations	Rectitude, respect, knowledge, skill, voice (power); collaboration (skill, respect)		
		Homeowners associations	Power, wealth, knowledge; collaboration (skill, respect)		
		Neighborhood organization	Knowledge, voice (respect); collaboration (skill, respect)		
		Firewise USA® and other fire safety organizations	Collaboration (skill, respect); knowledge (enlightenment)		
		Local businesses	Wealth, voice (power); collaboration (skill, respect)		
		Agrarians (farmers, ranchers, fishers)	Collaboration (skill, respect); technical skill		
		Academics and research organizations	Enlightenment, skill, respect, collaboration (skill, respect)		
		Insurers	Wealth, skill; collaboration (skill, respect)		
Social equity	Homeowners	Lower income	Family, Native nation affiliation, friends (affection, respect); neighborhood and community organizations (skill, enlightenment, affection); faith-based community (rectitude)	Reliance on and collaboration with family, neighbors, faith-based actors, social services, enabled by general agency budgets and targeted wildfire resiliency and disaster relief funds; sharing of information enabled by translation services and informal networks; self-help	Recognition for, justice for, and empowerment of vulnerable populations. Direct participation of marginalized individuals and communities in wildfire protection planning institutions. Incorporation of local and Indigenous knowledge in wildfire management strategies. Intergenerational wealth transfer through property ownership and inheritance. safe homes. Empowered, respected, resilient communities
		Senior citizen			
		Occupant of multi-family Dwelling			
		Racialized population			
		Native American			
		Documented immigrant			
		Undocumented immigrant			
		Limited English proficiency			
		Persons with lower level educational attainment			
		Persons with disability			
	Renters	See socioeconomic categories directly above	Friends (affection); informal networks (affection, enlightenment)	Reliance on and collaboration with friends and neighbors; sharing of information enabled by translation services and informal networks; access to rental assistance; self-help	
	Uninsured	See socioeconomic categories directly above	Family and friends (affection); informal networks (affection, enlightenment); community services (wealth, skill, enlightenment)	Reliance on and collaboration with family, neighbors, faith-based actors, social services, disaster relief, e.g., emergency shelter; sharing of information enabled by translation services and informal networks; self-help	
	Government	Social service agencies and firefighting services	Statutory and regulatory power; budget resources, physical resources (e.g., public housing); and knowledge (wealth, wellbeing, enlightenment); collaboration with other state and nonstate actors (rectitude, skill, respect, power)	Provision of public assistance and disaster relief enabled by general agency budgets and special allocations (e.g., declared emergencies); disaster preparedness and resiliency funds from federal and/or state government	
	Nonstate actors	Advocacy organizations for marginalized populations; nonprofit health and welfare organizations	Collaboration with nonstate and state actors (rectitude, skill, respect); knowledge of and access to relevant state funding (enlightenment); indigenous knowledge of wildfire (enlightenment; skill); budget resources and eligibility to administer state-funded aid (wealth, respect)	Help with installation of home hardening and defensible space measures; integration of traditional and official wildfire management strategies (skill; power); provision of emergency shelter, rental assistance; dissemination of information on funding sources for wildfire mitigation and resilience funds (skill; enlightenment; wealth; rectitude)	

The rational actor, governance, and social equity literatures mostly present the insurance industry as a homogenous actor without differentiating among different industry participants. This is a notable gap across the literatures, considering the variety of actors and functions performed—for example, brokers (who represent purchasers (consumers) of insurance policies), agents (who represent the insurance companies that employ them), admitted insurers (licensed in the state where they operate), surplus line insurers (licensed outside the state where the insured party resides), and reinsurers (companies that insure the risk of other insurance companies). As traditional insurance companies become more circumspect about insuring properties in wildfire-prone areas, emerging and lesser known actors in the industry (e.g., companies that combine sophisticated risk-forecasting services and specialized insurance products) will become increasingly important participants in insurance and reinsurance markets. This has implications for models and methods used in rational choice research, in governance experts' assumptions about cross-sectoral collaboration, and in analyses of vulnerable populations' access to insurance.

From a Lasswellian policy sciences perspective: all wildfire management actors are motivated by sought-after goals. These goals, and the strategies for achieving them, depend on participants' value dispositions. Lasswell made reference to the same set of eight values that people use as a foundation ("base values") for accumulating other desired goals and end states ("scope values"). These values are power, enlightenment, wealth, wellbeing, skill, respect, rectitude, and affection. It is important to note that Table 2 does not provide comprehensive accounts of either value assets (base values) or value demands (scope values) for all key participants. Rather the objectives are to a) highlight the values emphasized in the three categories of literature, b) draw attention to participants' dominant strategies, and c) identify the preferred end states of the collective suite of participants. The collective perspective is important, according to Lasswell, because it reflects the "common interest" in any given policy context (Clark, 2002: 13–15). Hence, the rational actor literature presents homeowners as actors seeking to maximize economic return (Essen et al., 2023). Homeowners choose to abate or not abate, or buy or decline insurance, based on a rational calculus of benefits, risks, and costs (Boomhower et al., 2023; Lankoande et al., 2005; McKee et al., 2004). *Wealth* is the key scope value for homeowners as well as for insurers, though homeowners seek to preserve wealth through an insurance contract, whereas insurers generate wealth through actuarial *skill* and knowledge (*enlightenment*). The irreducible object of study is the homeowner in the rational actor literature, and hence, generalizations about how markets behave are predicated on theoretical models and/or the risk-seeking and risk-avoiding behavior of experimental subjects or survey respondents.

In contrast, multi-scale, multi-level, networked sets of actors, chains of authority, flows of information and funding, and learning and decision pathways are key constructs in the governance literature. The valuable assets generated by these collaborative and/or interdependent sets of actors whether organized through polycentric (Abrams et al., 2021; Bixler et al., 2023), co-managed (Nowell et al., 2022), or coupled natural/human (Fischer et al., 2016) systems often revolve around technical, bureaucratic, and experiential information (*enlightenment*), sets of interconnected rules (*power*), coordinated roles (*skill and respect*), relationships of trust (*respect* and *rectitude*), and colleagueship (*affection*) in wildfire management regimes (Essen et al., 2023). Preferred collective outcomes, like resilient communities, are sought-after end states in the governance literature. In successful regimes, key participants value collaboration, which implies both skill and respect. Collective action is a distinctive feature of governance systems—in marked contrast to individual, rational actors making exclusive, independent decisions (Essen et al., 2023).

Whereas optimal sharing and use of information by multiple parties, nimble coordination, and ability to pivot in response to external cues are essential elements of strategy in the governance literature, coping strategies are more salient in social equity scholarship. Family (*affection*) is essential to vulnerable communities (e.g., family as repositories for information and as purveyors of indigenous and traditional knowledge of wildfire management; reliance on family for shelter in instances of property loss). Insofar as family are important to disadvantaged communities, this may reflect the failure of markets and governmental institutions to provide adequate services. Nevertheless, public agencies responsible for social services and community-based nonprofit organizations are key participants in the social equity literature and to some extent, the governance literature (Auer, 2021). Networks are featured in both types of research. Hence, development of a wildfire hazard mitigation program for a lower income community, involving neighborhood-based organizations and local fire departments, may be supported by funding from a FEMA grant, and a municipality's building codes may be inspired by norms adopted by a homeowners association. As Table 2 shows, preferred outcomes for the collective set of key actors in the social equity literature include, but are not limited to, wildfire safety and property protection. Being included and having a stake in institutions for wildfire management engage values such as power, enlightenment, skill, respect, and rectitude.

Conclusion

A research agenda that integrates concerns about homeowners insurance, wildfire governance, and impacts of wildfire on vulnerable populations is timely—even urgent. That view is supported by worrisome trends, including shrinking markets for affordable insurance in wildfire-prone states (Quinton, 2019; Walters, 2021), and the need for problem-solving governance institutions that can alternately fortify insurance markets and/or provide insurance services amid market failure. This review of relevant literatures finds divergent perspectives on key questions governing wildfire and homeowners insurance. Lasswell's social process framework provides a tool for ordering and categorizing the disparate human, social, and institutional dimensions of wildfire management and the role of insurance. The social process overlay reveals significant differences across the literatures in terms of the people, perspectives, and institutions that participate in wildfire risk management. Rational actor-oriented research considers the predicted or revealed behavior of comparatively limited sets of key actors who respond to well-defined, frequently unidirectional flows of information. This contrasts with governance studies that present wildfire as complex social-ecological problems requiring collaborative regimes and sophisticated institutions for information-sharing. The social equity literature illuminates the plight of marginalized actors—a challenging perspective for the rational actor framing, since some vulnerable populations may be unable to self-protect or buy insurance in the first place.

Behavioral experiments, preferred by some rational choice experts, would be strengthened by attending to the intersection of race, income, housing instability, limited options for insurance, and other variables that otherwise complicate assumptions about rational choice. At the broadest level, the three, dominant research themes can learn from one another, and integration of perspectives, epistemologies, and methods could generate insights (diagnostic and prescriptive) that are otherwise missing in current research programs (Miller et al., 2008).

Well-traveled research pathways on mitigation versus insurance should make room for complexity and socio-economic variability. Complexity is the domain of governance experts, and class and economic segmentation are core subjects in social equity scholarship. There is also room for applications of governance thinking that addresses the plight of vulnerable populations. To its credit, the National Cohesive Wildland Fire Management Strategy (2014: 18–19) recognizes "disadvantaged communities" and "social vulnerability" as categories for risk identification. Less well understood are demonstrated, replicable governance practices that foster wildfire resilience among vulnerable communities.

Social and behavioral scientists studying wildfire risk and insurance also need to explore new types of government and market-oriented interventions. Consider, for example, recently enacted or proposed regulatory changes in California meant to protect policyholders, including periodic moratoria on insurance non-renewal in/near declared wildfire disaster zones and mandates on insurers to provide discounts for homeowners who mitigate risks. The governance literature should attend to prospects of policy diffusion to other states. Will California provide a model in this space as it has for other environmental problems? What are plausible sources of resistance to these innovations and what kinds of governance structures and practices can overcome obstacles to make homes safer from destructive wildfire?

Even as California has adopted measures to protect homeowners, there is increasing pressure on that state's lawmakers to give insurance companies more discretion in rate-setting (Singh 2022). Rates in many states have increased steeply over the past few years. Allowing insurance companies to set rates based on estimates of future losses, rather than on past losses, could pose new challenges for lower income homeowners. This is an important topic for social equity experts to consider. Meanwhile, prospective remedies for rising insurance costs, like community-embedded insurance, deserve attention from governance experts, since these insurance products may require collaboration among community-based, private, and public entities.

Rational choice researchers could more clearly define the full suite of challenges to insurance markets and to insured and uninsured populations as insurance of last resort becomes increasingly common. In addition, more research is needed on the promises and potential pitfalls of emerging sources of government aid, including for wildfire resilience funds (e.g., under the 2021 Infrastructure Investment and Jobs Act and the 2022 Inflation Reduction Act)—a subject for social equity experts to consider, in particular.

All three categories of literature share one attribute: a tendency to view insurance as a notional or abstract feature of wildfire management. It is assumed that insurance exists in the marketplace—that it is available (though it may not be affordable or easily accessible). This conceptualization of insurance may need to change if insurance becomes increasingly scarce or ceases to exist in its traditional forms. These prospects are real, considering the continuing perils of climate change and the skittishness of insurers in the most risk-prone areas of the country. Many of the contributions to the literature reviewed in this article recognize climate change as a major driver of wildfire risk. Comparatively few make reference to the inverse relationship between climate change-fueled wildfire risk and availability of high quality, affordable insurance. Students of the Lasswellian tradition may prove especially adept at correcting this research shortcoming considering the policy sciences' strengths in mapping changes in trends and conditions and in devising policy alternatives in the common interest.

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